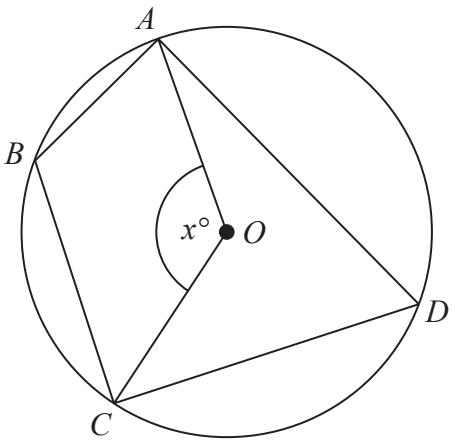
13 A, B, C and D are points on a circle, centre O.



Find, in terms of x, the size of angle ABC.
Give a reason for each stage of your working.

Angle ADC = x/2

Angle at the centre is twice the angle at the circumference on the same arc.

1 mark
Use the arc AC that does not contain D.

Angle ABC + angle ADC = 180°

Opposite angles of a cyclic quadrilateral sum to 180°.

Angle ABC = 180° – x/2

1 mark
Subtract the opposite angle.

1 mark
Both circle-theorem reasons are linked to their steps.

180 – x/2 °

(Total for Question 13 is 3 marks)

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